

the Cartan involution θ . Furthermore as part of the same theorem one has that θ can always be chosen in such a way that it maps the compact Cartan subalgebra into itself:

$$\theta : \mathcal{H}_c \rightarrow \mathcal{H}_c \quad (1.20)$$

This short discussion reveals that the classification of real sections of a complex Lie Algebra $\mathfrak{g}$ is in one-to-one correspondence with the classification of symmetric spaces, the complexification of whose Lie algebra of isometries is $\mathfrak{g}$.

Let us now consider the action of the Cartan involution θ on the Cartan subalgebra:

$$\mathcal{H}_c \equiv \text{span}\{iH_j\} \quad (1.21)$$

of the maximal compact section $\mathbb{G}_{\text{comp}}$. The subspace of $\mathcal{H}_c$ belonging to the eigenspace of θ corresponding to eigenvalue 1 is, according to eq.(1.15), part of the subalgebra $\mathbb{H}_\theta = \mathbb{H}_c$, while the subspace of $\mathcal{H}_c$ belonging to the eigenspace corresponding to the eigenvalue -1 is, according once again to eq. (1.15), part of the subspace $\mathbb{P}$. It follows that the Cartan subalgebra of the real section $\mathbb{G}_\theta$ can be split in two subalgebras as follows:

$$\begin{aligned} \mathbb{G}_\theta \supset \underbrace{\mathcal{H}_\theta}_{\text{CSA of } \mathbb{G}_\theta} &= \mathcal{H}_\theta^{\text{comp}} \oplus \mathcal{H}_\theta^{\text{n.c.}} \\ \mathcal{H}_\theta^{\text{comp}} &\equiv \mathcal{H}_c \cap \mathbb{H}_\theta \\ \mathcal{H}_\theta^{\text{n.c.}} &= i\mathcal{H}_c \cap \mathbb{P}_\theta \subset \mathbb{K}_\theta \end{aligned} \quad (1.22)$$

Since, as stated by the quoted theorems, any real section $\mathbb{G}_R$ of a complex $\mathfrak{g}$ is isomorphic to $\mathbb{G}_\theta$ for a suitable θ , it follows that the splitting of the Cartan subalgebra of any $\mathbb{G}_R$, into a compact and non-compact subalgebra has an intrinsic meaning. Hence for every real section $\mathbb{G}_R$ we can define the compact and non-compact rank as follows:

$$r_{\text{n.c.}} = \dim \mathcal{H}_\theta^{\text{n.c.}} \leq r \quad ; \quad r_c = \dim \mathcal{H}_\theta^{\text{comp}} = r - r_{\text{n.c.}} \quad (1.23)$$

Having clarified the relation between real sections $\mathbb{G}_R$ of a complex Lie algebra $\mathfrak{g}$, Cartan involutions θ and non-compact Riemannian symmetric spaces $\frac{\mathbb{G}_\theta}{\mathbb{H}_\theta}$, let us then reconsider a symmetric coset manifold:

$$\mathcal{M}_{\mathbb{G}_R} = \frac{\mathbb{G}_R}{\mathbb{H}_c} \quad (1.24)$$

where $\mathbb{G}_R$ is a simple Lie group and $\mathbb{H}_c \subset \mathbb{G}_R$ is its maximal compact subgroup. The Lie algebra $\mathbb{H}_c$ of the denominator $\mathbb{H}_c$ is the maximal compact subalgebra $\mathbb{H}_c \subset \mathbb{G}_R$. Denoting, as usual, by $\mathbb{K}$ the orthogonal complement of $\mathbb{H}_c$ in $\mathbb{G}_R$:

$$\mathbb{G}_R = \mathbb{H}_c \oplus \mathbb{K} \quad (1.25)$$

we define **rank of the coset** $\mathbb{G}_R/\mathbb{H}_c$ the dimension of the non-compact Cartan subalgebra:

$$\text{rank}(\mathbb{G}_R/\mathbb{H}_c) = r_{\text{nc}} \equiv \dim \mathcal{H}^{\text{n.c.}} \quad ; \quad \mathcal{H}^{\text{n.c.}} \equiv \text{CSA}_{\mathbb{G}_R} \cap \mathbb{K} \quad ; \quad \mathcal{H}^{\text{comp}} \equiv \text{CSA}_{\mathbb{G}_R} \cap \mathbb{H}_c \quad (1.26)$$

Next, choosing a basis of $\mathcal{H}_c$ aligned with the simple roots:

$$\mathcal{H}_c = \text{span}\{iH_{\alpha_i}\} \quad (1.27)$$

we see that by duality of vector spaces the action of the Cartan involution θ is transferred from the Cartan subalgebra to the simple roots α_i and hence to the entire root lattice. As a consequence we can introduce the notion of real and